

Accomplishing More With Less: The Practice of Cybersecure Health Technology Design Among Danish Startups

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ABSTRACT

While innovative health technologies have the capacity to better lives, their success hinges, increasingly, on their cybersecurity. Recent years have seen growing awareness of cybersecurity as not only technical but cultural in nature. And yet, how health technologies are developed is also changing — innovation today often led by startups, about whose cybersecurity efforts we know less, and whom might be most at risk. Striving to address this essential gap in knowledge, we present findings from a mixed-methods, design-led study of nine Danish health technology startups and six cybersecurity experts' experiences. Through analysis of interviews and workshops, we contribute insight into startups' perceptions of cybersecurity as essential yet often elusive, one amongst many priorities, and often a self-taught discipline. Drawing on this knowledge, we describe the co-design of a self-assessment platform to support increased awareness and preparedness among health technology startups, concluding with implications for future cybersecurity initiatives.

CCS CONCEPTS

• **Security and privacy** → **Human and societal aspects of security and privacy**; • **Human-centered computing**;

KEYWORDS

Cybersecurity, Human-Computer Interaction (HCI), design, development, health technology, startups, enterprise software

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1 INTRODUCTION

New and innovative health and healthcare technologies have the capacity to change our lives for the better. And yet their success appears to hinge, increasingly, on their cybersecurity. In 2022 alone, the sensitive health data of 9.7 million Australians was compromised and hundreds had their data published online [5]. The U.S. Food and Drug Administration provided new warnings about medical device vulnerabilities risking direct harm to patients [25] and two-thirds of healthcare organisations surveyed worldwide reported that they were subject to ransomware attack [58] — one single instance causing the shutdown of all major information technology systems of the Irish national health service [54].

Cybersecurity concerns are therefore of increasingly critical importance for health technology developers. While technological progress is fast expanding that which is possible in healthcare, this has equally rendered our data - and ourselves - increasingly vulnerable; to sensitive personal data leaks [15], impersonation [5], disruption of critical services, false information delivery [36], and perhaps even death [20]. And yet, cybersecure health technology design is itself an increasingly complicated and multi-faceted task requiring deep expertise in domains as broad as health data standards, cybersecurity best practices, and knowledge of users' behaviour.

The complex nature of these challenges has recently led leading cybersecurity researchers to describe the problem of cybersecure health technology development as substantially 'cultural' in nature [20]. And while much Computer Science research has focused on the development of cybersecurity standards [8, 22, 23] and guidelines for practice [53], less research has focused on the humans at the heart of development, and increasingly the source of vulnerabilities exposed by malicious actors, negligence and accident [33, 68].

How health technologies are developed is also changing however — innovation often driven by startups, about whose efforts to manage, prioritise and respond to cybersecurity concerns we know less, and whom might be most at risk given the vast and complex nature of this domain [36, 39], the diversity of organisational structures,

and increased dependence on external services such as the cloud [37, 52]. To develop our knowledge of not only the theory but practice of cybersecurity, we are required to understand whom takes responsibility for cybersecurity in small teams, how they remain informed about guidelines and regulations, which methods they employ, and how these efforts interface with other critical development tasks. The question then also arises, just how might we best leverage such insight in support of the design of tools, methods and materials to support increasingly effective cybersecurity practices?

To this end, we present in this paper, findings from a mixed-methods, open-ended design study engaging six cybersecurity experts and 17 executives, founders and engineers of nine Danish health technology startups in interviews and workshops for the purpose of obtaining insight into their experiences of developing cybersecure health technologies. Through this research, we contribute new knowledge of startups' perceptions of cybersecurity as essential yet often elusive, one amongst many priorities, and often a self-taught discipline. We then employ this insight to inform the co-design of a novel *Self Assessment Platform* to support increased cybersecurity awareness and preparedness among health technology startups, concluding with implications for the future design of cybersecure health technologies.

2 RELATED WORK

Each of the fields of Medicine, Computer Science and HCI today contribute unique perspectives on the key challenges of cybersecure health technology development.

Cybersecurity as Medical Data Regulation Problem. From ever new AI-assisted diagnostics, healthcare and telehealth systems [34, 39] to ever more connected pervasive and Internet-of-Things (IoT) technologies [50, 64], digital health tools are often today described as promising improved well-being for us all. And yet, this future is increasingly challenged by cybersecurity concerns [56, 64]. For health professionals in particular, enthusiasm for new technology is often tempered by fears about incompatibility [29, 41], responsibility for critical functions [28, 29], and the potentially disastrous consequences of malware, data breaches and denial of service attacks for patients' health [20, 41, 62]. Increased medical and health data regulations, from the EU's GDPR [8] to HIPAA [27] are seen by many as a solution to these concerns; improving oversight, minimising patient risk and increasing data security [8, 22, 42]. Yet others however, see standardisation as an inhibitor of innovation [23, 27], decrying a one-size-fits-all regulatory approach [23] in favour of guidelines and the development of new cybersecurity tools [53]. There thus persists in the medical and legal fields a lack of clear consensus around just how to best protect users and systems.

Cybersecurity as Development Problem. The pressing nature of the cybersecurity threat has yet led many computer scientists to turn to the development of models of cybersecurity maturity, best practice guidelines and new processes [10, 35, 36], including secure authentication, authorisation, and data access control mechanisms [43, 44, 53], as means to both increase security and reduce costs [39, 66], by decreasing the siloing of developers and cybersecurity departments [6] and establishing shorter development cycles for example. Many expert practitioners have thus emphasised the need

to see not only the use-in-practice of digital health technologies as key sources of cyber-insecurity, but also the processes of their development [21, 42, 61]. There is much we have learned about cybersecurity best practices in this respect, and yet new health technologies prove still too often detrimental to the well-being of their users as a result of inadequate cybersecurity measures [2, 51, 60].

Cybersecurity as Design Problem. To arrive at increasingly cybersecure health technologies, prior research then suggests, we must equally attend to the central role of the many key stakeholders in their development and use. Initial mixed-methods research efforts in this regard have focused on developing insight into developers' practices [57, 63, 67], including through online surveys of software developers [3]; drawing attention to the importance of developing a cybersecurity-aware culture with the buy-in of end-users [1, 24, 30], of aligning the needs and values of engineers and users [8, 18, 40], of encouraging developers to create health technologies that are not only secure but do not compromise users' experience [11, 26], and of protecting particularly vulnerable user groups [9, 49, 59].

Here there lies also a new opportunity for HCI researchers and practitioners to create value, through the participatory co-design of novel solutions to these long-standing and complex human problems – cybersecurity itself often a question of the design of appropriate supports [12, 22, 38]. To date researchers have employed mixed-methods, expert interviews and participatory workshops to support the design of other tools, including for example secure smart home devices [7, 11, 12], indicating that the diverse participation of cybersecurity practitioners, developers and end-users can yield improved cybersecurity solutions for all [11, 12, 63]. And yet, while much research has focused on identifying problems in practice, less has focused on the design of solutions to support cybersecure health technology development [3], particularly with startup companies, and the health technology context in mind [31, 63].

3 METHOD

Our aim in this work was then to obtain insight into Danish health technology startups' experiences, needs, methods, tools and processes for mitigating, attending to and responding to cybersecurity considerations – with the parallel goal of leveraging this insight to inform the design of new tools, methods and/or materials for a security-conscious approach to health technology development. To this end, we engaged six cybersecurity experts and 17 executives, founders and engineers at nine Danish health technology startups in interviews and workshops – a mixed-methods approach to design research we structured following a blended Double Diamond and SCRUM framework composed of six sprints (See Fig 1).

During the *first sprint*, we engaged 12 founders, engineers and executives of eight health technology startups spanning mobile health apps, medical AI diagnostics and medical devices in eight interviews exploring health technology startups' cybersecurity experiences (See Table 1). Startups were recruited through email messages sent to the Technical University of Denmark, a local startup organisation, and an international startup organisation, individual startups, and the authors' professional networks.¹ All participants were provided

¹ 57 startups were contacted for interview, 28 offering digital health solutions, 23 other medical technologies, and six other medical tools and systems.

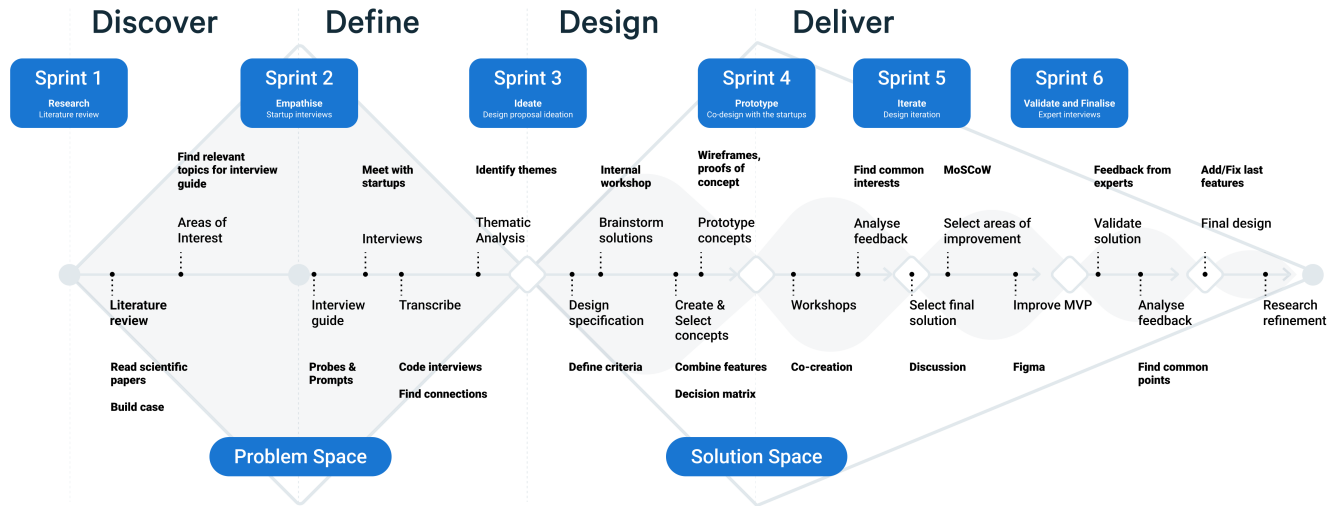


Figure 1: Our Design Research Approach

an information sheet in advance of their involvement communicating the anonymous handling of their data in accordance with best data management practices, GDPR and university regulations, in addition to their right to withdraw from the project at any point in time. Each interview was conducted by the first two authors in either English or Danish, employing a semi-structured interview guide covering eight key areas of interest (See Appendix A). Interviews lasted approximately one hour; three conducted online and five at the startups' offices. Nine hours of recorded audio were subsequently transcribed in full, all identifying information redacted, and the recordings deleted.

During the *second sprint*, these transcripts were then analysed in their respective languages using ATLAS [4] and reflexive thematic analysis – interviews divided between the first two authors and coded separately in English following Braun and Clarke's guidance [13], codes then organised into themes, and refined with the involvement of the last author through an approach both inductive and deductive in nature. The *third sprint* began with the creation of a design specification informed by the results of this analysis, and was followed by a series of internal workshops comprising problem space convergent activities including the visualisation, reframing and categorisation of problems, brainstorming [45] and mind-mapping [47] exercises (See Appendix B). These activities led to a multitude of creative concept solutions to the task of supporting security-conscious health technology development, of which the three most promising were chosen with the aid of a weighted decision matrix [46] for initial wireframing, followed by higher-fidelity prototyping (See Appendix C).

The *fourth sprint* then entailed the conduct of co-design workshops with nine participants representing four startup companies for the further development and validation of these concepts (See Table 1). Each workshop lasted two hours: two conducted online, one in-person and one in hybrid form using a combination of paper-based templates, post-it notes and the whiteboard platform Miro [48]. During workshops, participants were presented with both a summary of findings from the initial interview phase and the

concept solutions in-development, and then invited to engage in further discussion, brainstorming and testing of prototypes; the research team gathering feedback both qualitative and quantitative in nature through activities including the ranking of problems and dot voting [19]. These insights next informed the selection and refinement - during the *fifth sprint* and by means of the MoSCoW method [14] - of the single concept deemed most feasible and likely to provide greatest value to the startups.

For the *sixth sprint*, this refined solution was evaluated through a series of interviews conducted with six senior cybersecurity experts (E1: Head of Cybersecurity Organisation, E2: Consultant at Cybersecurity Firm 1, E3: Software Developer Specialised in Cybersecurity, E4: Security Architect, E5: Software Developer at Cybersecurity Firm 2, and E6: Cybersecurity Consultant and Startup Founder), entailing think-aloud testing of the prototype, and completion of a verbal questionnaire (See Appendix A). These expert participants were recruited through prior professional networks and connections facilitated by the participating startups. A topic summary of the notes made during these expert interviews informed a final iteration of the prototype solution.

4 FINDINGS

We next report findings from analysis of our interviews, revealing cybersecure health technology design as considered essential yet often elusive, one amongst many priorities for startups, and often a self-taught discipline.

4.1 Cybersecurity is considered essential yet often elusive in practice

Cybersecurity was described by participants across these interviews as many things, with many influences – as defined by certification for example, from ISO27001 for information security to ISO13485 for the CE marking of medical devices, and yet principally technical in nature; “[the] technical implementation of things that ensure we are not breaching anything” (P4). Network security

Company	Interview Participants	Workshop Participants
Digital Health App Developer	[I1] Project Lead (P1)	-
Medical AI Diagnostics	[I2] Clinical Operations Director (P2)	-
Medical AI Diagnostics	[I3] CTO (P3)	-
Medical AI Diagnostics	[I4] CTO (P4), Project Manager (P5), Software Developer (P6)	[W2] Project Manager (P5), Software Engineer (P6)
Medical Devices	[I5] CTO (P7), CIO (P8)	[W1] CEO (P13), CTO (P7), CIO (P8)
Medical Devices	[I6] Founder (P9)	-
Medical Devices	[I7] CEO (P10), Engineer (P11)	-
Medical Devices	[I8] CTO (P12)	[W4] - CTO (P14), Software Engineer (P15), "IT guy" (self-described) (P16)
Telehealth Provider	-	[W3] CTO (P17)

Table 1: Participating Stakeholders (company details anonymised). I: Interview | P: Participant | W: Workshop

was in this regard the means of risk mitigation most frequently described, through application firewalls or isolating products within customers' networks. The CIO of one medical device company described continuous simplification as a key tenet of their approach; *"You could say our mission is to make it so simple that it becomes hard to mess up."* (P8). While many other participants spoke also of their company culture as equally critical to reducing hacking risks and data leaks — a culture of best practices in which *"even the sales personnel should make sure their passwords and usernames are hard to get"* (P3). The majority of startups then described recognising the importance of not only best practices and regulations including GDPR, but formalised processes for handling problems should they arise; organisation-wide strategies requiring the involvement of top-level management.

Cybersecurity was equally described by many however as not only about product-development or organisational best practices but a selling point for customers and a prerequisite for doing business. Many participating startups described customers as increasingly aware of standards including ISO27001, which they employed as an indicator of cybersecurity maturity and readiness; *"that is beginning to be requested. Back when we started they just asked, 'is your system secure?' and we just said 'yes' "* (P3). Depending on the company's size, this could be interpreted as either an opportunity or a roadblock; *"a ... parameter that distinguishes you from other organisations"* (P3). The participating startups in these interviews therefore demonstrated clear awareness of the importance of cybersecurity, for their products, customers, and business. All were aware of the dangers involved and actively worked to mitigate these through extensive risk assessments, audits and technical initiatives to mitigate both health data and intellectual property risks. And yet, divergent opinions also surfaced regarding many cybersecurity measures. The use of cloud services, for example, was often considered contentious; some promoting this approach as *"more secure than on-premise"* (P3), while others believed it less secure. Thus, these interviews furthermore revealed opportunities to clarify the multi-faceted and often context-specific nature of cybersecurity concerns both technical and cultural in nature.

4.2 Cybersecurity is one amongst many priorities for startups

Indeed, participating startups rarely spoke of their cybersecurity needs and concerns in isolation but as entangled with and complicated by many other aspects of startup culture and organisation.

One major source of risk in the development of cybersecure health technologies is therefore startups' capacity to prioritise privacy and security in an ever fast-paced and high-pressure environment. All participants spoke of research and development as their imperative priority, born of the unrelenting need to render their products and business successful. As the founder of one medical device startup put it; *"if there isn't money in it ... then you would not prioritise it, ... would you rather close or risk ... a cyber-attack?"* (P9). This was a problem compounded by pressure on time and resources, including in regards to the numbers and expertise of available personnel. Many participants spoke of turning to external help in the form of consultancy, and of attempting to make up for a lack of expertise through their active participation in workshops and talks hosted by a regional cybersecurity organisation, or by working to independently coordinate meetings with other startups and clients.

For many startups then, cybersecure health technology design becomes a case of doing more with less. This could manifest itself in many ways, as reflected for example in one startup's decision to limit the scope and functionality of a medical product, in order to *"develop and sell it as an electrical device"* (P12), so as to circumvent the resource-intensive task of CE marking. It equally became clear over the course of these interviews however that startups' need to focus on many other priorities often engendered a sense of insecurity about cybersecurity; which resources they really needed, and when they needed them. Knowing how and where to start in this regard was often described as one of the most difficult challenges for startups; *"It's inexperience ... Who do you find to do it? For the best price? Who is the most competent? So it's pretty much been, what do you do if you don't know what to do?"* (P9). Consistent expression of this need regardless of company size or product focus made for one of the most striking takeaways from these interviews.

4.3 Cybersecurity is often a self-taught discipline

For these participating health technology startups then, cybersecurity was less a well-formalised and understood series of steps than a gradual, iterative and ever-evolving process of developing knowledge and experience. Company scale did yet emerge as exerting a significant influence over startups' capacity to acquire sufficient internal competencies and resources, often necessitating increased outsourcing of cybersecurity-related tasks. Participating companies varied significantly in age and size, although approximately half had under ten employees, and the challenges facing these smaller

startups in particular hinged also to a significant extent on the background of their founders. Many participants spoke of the need to adopt multiple roles; *“when we started this, ... you had to do everything”* (P3), cybersecurity responsibilities often falling to those with the most technical of backgrounds, and typically the IT or software responsible; *“I got the classic one-man responsibility for both roles, ... cybersecurity ... of course the most relevant”* (P8).

This also meant that the most pressing challenges facing each startup would often simply depend on the competencies available to the team at that point in time. Many participants expressed drawing on knowledge acquired from previous workplaces, meaning that for one of the largest startups interviewed, ISO27001 certification was described as their most challenging task, while another newly-launched startup spoke of the benefit of an employee with experience certifying others to the same standard. The relevance of regulation was however also questioned by some participants, whom felt that it was difficult to create standards to suit startup companies of all sizes equally; *“you cannot take a template somebody else has made [and] run it into our own QMS [quality management system] ... smoothly”* (P3). For those startups whom worked with external healthcare partners, cultural differences and work practices often proved a source of frustration. Many spoke of hospitals’ conservative character as greatly impacting their ability to work and collaborate, *“at hospitals, if you can, completely avoid having to talk to the IT department”* (P12), and of contrasting expectations for cybersecurity and patient data awareness as exasperating; *“if you asked the individual clinicians, doctors, nurses ... they DON’T THINK about it at all. They have NO IDEA what it is”* (P10).

Startups demonstrated a highly-iterative approach to the development of their products, which they equally applied to their cybersecurity practices; learning from incidents spanning cloud storage problems and server misconfigurations, a pattern seen often across interviews. This meant that for many startups, cybersecurity became in many respects a ‘self-taught discipline.’ While a risk in itself, such realities equally present the opportunity to support health technology startups in securing their products and services through tools, methods and materials designed with their unique needs in mind.

5 DISCUSSION

We strove in this work not only to develop new knowledge of the needs and concerns of health technology startups but to leverage this insight in support of the user-centred design of a novel solution to meaningfully impact these companies’ ability to develop increasingly cybersecure health technologies.

5.1 Co-Designed Concept Solutions

The early stages of this work entailed the development of numerous concept solutions to the needs and challenges of the participating health technology startups (See Appendix B), narrowed down to three prototypes chosen and refined with participants’ input (See Appendix C). This included; (1) a web-based **Self Assessment Tool** through which completion of a short questionnaire covering users’ goals, time available, areas of expertise and confidence in cybersecurity domains, generates a ranking of recommended cybersecurity initiatives; (2) a **Course Base** comprising a learning platform to

support increased cybersecurity awareness through dissemination of traditional course materials as well as health technology-specific case studies reflecting the key challenges surfaced through this research; and (3) a **Business Toolkit** entailing a curated list of tools for health technology development - from project-management software to cloud service providers - facilitating at-a-glance comparison of the cybersecurity implications of these business-critical tools.

Further discussion and co-design of these concepts in collaboration with health technology startup founders, executives and engineers (See Section 3) led to the selection of the **Self Assessment Tool** as the concept solution deemed most novel and feasible, most valuable to new and smaller-scale startups due to its emphasis on accessibility, and with the greatest potential to impact practice. Continued development on the basis of expert feedback led to the addition of several new features, including a landing page and tutorial functionalities, revised dashboard and self-assessment features, an expanded knowledge base and opportunities for achievements (See Fig. 2 & Appendix D). Although other tools to support the adoption of cybersecurity initiatives among smaller organisations do already exist [16, 17, 32, 55, 65], this particular solution - the result of an interdisciplinary and mixed-methods approach - provides an interactive experience grounded in the key challenges facing health technology startups, actively encourages team members to continually improve their startups’ cybersecurity through gamified features including badges and leaderboards, and provides tailored visual representation of actionable initiatives based on individual startups’ needs.

5.2 A Tool to Support Increasingly Security-Conscious Health Technology Development

The aforementioned expert evaluation of this concept surfaced five key areas of opportunity to support the future development and adoption of this and other tools for security-conscious health technology development. First, experts were enthusiastic about the possibilities for **non-linear navigation** provided by the tool, a change inspired by workshop participants’ desire to ‘play around with’ and explore cybersecurity initiatives. Second, startups themselves likened the overarching functionality of the tool to the **familiar practice** of contacting external consultants for help, while experts noted that cybersecurity is not an entry-level discipline and that even though the tool was simple to use and understand, more could be done to make it more approachable for complete beginners. Third, all stakeholders praised the **tailoring** permitted by the tool, requesting also the option to strive for compliance with a specific region or country rather than individual regulations. Fourth, the tool’s user interface and dashboard received the highest praise, one cybersecurity consultant praising in particular the **time and difficulty estimates** provided, as permitting startups to efficiently select appropriate initiatives. And finally, opinions were most split on the addition of gamification, which the CIO of one medical device startup described as *“the driver of engagement”* (P3), while experts noted could also hold negative connotations (E2) and could better function if focused on self-improvement rather than competition (E3).

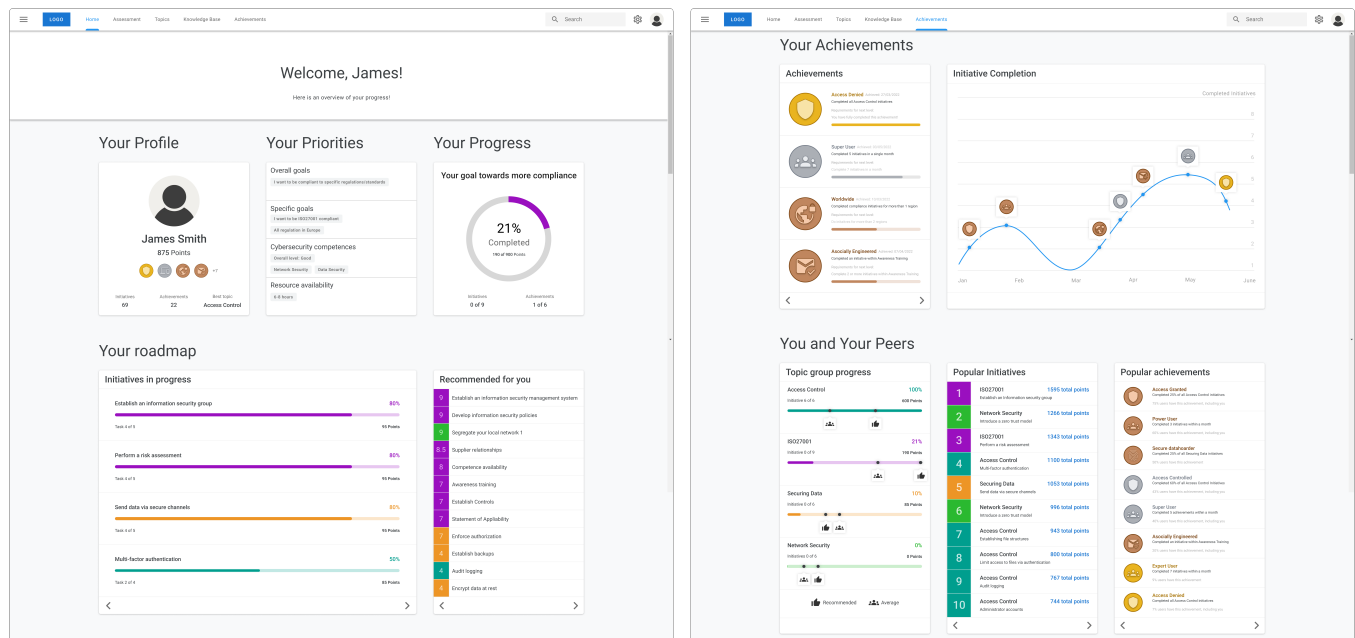


Figure 2: Self Assessment Tool | Left: Dashboard Page | Right: Achievements Page

5.3 Implications for Cybersecure Health Technology Design

Finally, we conclude this paper with a series of brief reflections on and implications for the practice of design to support increasingly cybersecure health technology development;

- **Simplicity of design and navigation makes possible tailoring to the diverse needs of individual startups**, which might in future iterations of this tool be supported by additional assessments of, for example, cybersecurity maturity.
- **Beware the potential for adverse misinterpretation of informal frameworks in a regulation and compliance heavy context.** The *Self Assessment Platform* currently employs a simple, non-formalised model for best supporting users according to their needs. It is essential however to mitigate potential conflation of this simpler model as itself permitting assessment of compliance with ISO or IEC standards, which are furthermore copyrighted and inaccessible without payment.
- **Design to support collaboration despite IP-intensive considerations.** Cooperation between startups and the health sector was highlighted across interviews as a particular pain-point in regards to cybersecurity, an often critical task which could well be better supported through future iterations of tools such as this.
- **Support the distribution of knowledge across diverse and flexible teams.** Founders, executives, engineers and experts spoke of the implementation of cybersecurity initiatives, and learning about them, as most effective in teams. This is a feature which future tools and systems would do

well to implement, through, for example increased integration with popular project management software, communication tools, and workflows.

6 CONCLUSION

The work of this paper has contributed new insight into health technology startups' cybersecurity experiences, informing in turn a novel, co-designed *Self Assessment Platform* for increased cybersecurity awareness and preparedness.

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